

Credit Card Compliance Platform

AI-Powered Regulatory Compliance

Regulatory analysis · Review workflow · Conflict detection · Live monitoring

UDAAP	TILA / Reg Z	ECOA / Reg B	FCRA	BSA / AML
PCI DSS	SCRA	FDCPA	SR 11-7	

Powered by Anthropic Claude · RAG Engine · ChromaDB · FastAPI · github.com/avika124/Card_Compliance

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1. The Problem & Solution Overview

Compliance is broken for most credit card teams — manual, slow, reactive, and with no institutional memory of prior commitments. The Compliance Platform solves all four problems in one system.

Manual & slow	Email-based reviews take days. No system of record.	No memory	Prior promises invisible when policies change.
Reactive	Changes found in exams, not when rules publish.	No audit trail	Who approved what? Nobody knows. Examiners ask.

Solution — four pillars

01 AI Analysis	Claude checks every document against 9 regulatory frameworks and returns findings with exact CFR citations. RAG ensures regulation text is retrieved before every check.
02 Company Memory	Prior marketing, policies, and scripts stored in a vector DB. Every new document is auto-checked for contradictions with prior commitments.
03 Review Workflow	Submit → Pending → Compliance Review → Decision. Role-based access for submitters, compliance officers, legal, and admins.
04 Regulatory Monitor	Watches CFPB, Fed, FFIEC, OCC, and PCI SSC. Alerts via in-app, Slack, and email the moment rules change.

2. AI Compliance Engine — 9 Regulatory Frameworks

Claude analyzes every document using Retrieval-Augmented Generation — relevant regulation text is retrieved from ChromaDB before each call, so findings cite exact CFR sections, not just framework names.

Framework	Full Name	Key Areas
UDAAP	Unfair, Deceptive, Abusive	Marketing, fees, customer harm
TILA / Reg Z / CARD Act	Truth in Lending Act	APR, billing, credit card terms
ECOA / Reg B	Equal Credit Opportunity	Adverse action, fair lending
FCRA / Reg V	Fair Credit Reporting	Accuracy, dispute rights
BSA / AML / OFAC / CIP	Bank Secrecy Act	AML, sanctions, KYC
PCI DSS	Payment Card Industry DSS	Data security, encryption
SCRA	Servicemembers Civil Relief	Military rate cap, protections
Collections / FDCPA	Fair Debt Collection	Contact rules, prohibited acts
SR 11-7	Model Risk Management	Model validation, oversight

Sample findings output

Each finding includes severity, regulation, excerpt from the document, citation, and a remediation recommendation.

Severity	Regulation	Issue	Citation
HIGH	TILA / Reg Z	APR missing from promotional headline — required when specific rate advertised	12 CFR 1026.16(b)
MED	ECOA / Reg B	Adverse action notice missing required anti-discrimination statement	12 CFR 1002.9
MED	FCRA / Reg V	Adverse action notice states 30-day dispute window — FCRA requires 60 days	15 U.S.C. §1681m
PASS	SCRA	Servicemember interest rate reduction procedure correctly documented	50 U.S.C. §3955

3. Company Memory & Conflict Detection

Every new submission is checked against your company's own prior communications — catching contradictions that external regulation checks miss entirely. Prior materials live in a ChromaDB vector store, and the most semantically relevant chunks are retrieved and sent to Claude alongside each new document.

What gets stored vs. What conflicts are detected

- **Marketing emails** — campaigns, offers, promotions
 - **Cardholder agreements** — T&Cs, pricing, amendments
 - **Internal policies** — credit, collections, underwriting
 - **Call center scripts** — onboarding, disputes, collections
 - **Settlements** — hardship agreements, consent orders
 - **Disclosures** — adverse action, change-in-terms notices
- **Rate & fee changes** — different APR or fees than prior
 - **Rewards terms** — earning/redemption differs from ads
 - **Product benefits** — described differently across materials
 - **Eligibility criteria** — qualifications inconsistent
 - **Legal rights** — dispute or arbitration language differs
 - **Explicit promises** — prior commitment contradicted

Example conflict detected

NEW DOCUMENT SAYS	PRIOR POLICY SAYS
"Earn unlimited 1.5% cash back — rewards never expire as long as your account is open."	"Unredeemed points are forfeited immediately upon account closure, with no grace period."
Q3 2025 Freedom Unlimited Email Campaign	Rewards Program Agreement v2.3 — CONFLICT: UDAAP unfair practice risk

Adding documents to company memory

Method	Description
Web UI upload	Drag-and-drop file (PDF, DOCX, TXT) with metadata: name, type, product, date, version
Paste text	Paste any text directly — marketing copy, policy sections, script content
URL scrape	Enter a URL — the platform fetches and indexes the page automatically
Bulk load	Load an entire directory of files in one click — included test docs or your own archive
CLI / API	POST /memory/add/file or /memory/add/text — fully automatable for batch ingestion

4. Review Workflow & Role-Based Access

Every document moves through a structured pipeline with full status tracking, decision logging, and multi-channel notification. No email threads, no lost approvals, no ambiguous status.

Workflow stages

1 Submit	Authorized user submits via web UI, file upload, or API. Captures title, product, channel, doc type, and priority.
2 AI Analysis	Claude runs compliance and conflict checks simultaneously. 9 frameworks + company memory queried. Results in <60 seconds.
3 Review Queue	Compliance officer reviews findings sorted by risk. High-risk auto-flagged. False positives can be marked and excluded from scoring.
4 Decision	Reviewer selects Approve / Reject / Escalate to Legal / Mark In Review. Notes required for rejection and escalation. Decision timestamped and attributed.
5 Notification	Submitter receives in-app alert, Slack message, and HTML email with the decision, reviewer notes, and link to full findings report.

Role-based access matrix

Role	Submit	Review Queue	Approve / Reject	Analytics	Settings
Admin	✓	✓	✓	✓	✓
Compliance	✓	✓	✓	✓	—
Legal	✓	✓	✓	✓	—
Submitter	✓	—	—	—	—

Submission statuses

Status	Meaning
Pending	Submitted, AI analysis complete, awaiting reviewer
In Review	Reviewer has opened and is evaluating the submission
Approved	Compliance officer approved — document cleared for use
Rejected	Rejected with required notes — submitter notified with reason
Escalated	Sent to legal review — triggers escalation notification

5. Regulatory Change Monitor

A background daemon fetches content from government websites on a configurable schedule (default: every 24 hours). Content hash comparison detects changes. When a change is found, all compliance, legal, and admin users receive multi-channel alerts before the next exam cycle.

Sources monitored by default

- **CFPB** — Reg Z, Reg B, Reg V, FDCPA, UDAAAP Guidance, SCRA Guide
 - **Federal Reserve** — SR 11-7 Model Risk Management
 - **FFIEC** — BSA/AML Examination Manual, CIP Requirements
- **OCC** — Comptroller's Handbook, Consumer Compliance Bulletins
 - **PCI SSC** — PCI DSS v4.0 Standards, Security Bulletins
 - **FDIC** — Financial Institution Letters, Compliance Guidance

Alert channels when a change is detected

Channel	What gets sent
In-App Notification	All compliance, legal, and admin users notified immediately with direct link to changed source
Slack Message	Formatted block message — regulation name, source, and action link to run an impact check
Email	HTML email with regulation, source name, and call-to-action to review affected materials
Audit Log	Change event recorded with timestamp, source URL, and content hash for examiner review

CLI commands

```
python -m monitor.reg_monitor --seed 'Acme Financial' # seed default watches
python -m monitor.reg_monitor --run-once # one check cycle
python -m monitor.reg_monitor --daemon --interval 24 # run every 24h
python -m monitor.reg_monitor --list # list all watches
python -m monitor.reg_monitor --delete-source 'Name' # remove a watch
```

Adding custom sources

Any URL can be added as a custom regulatory watch — state banking department guidance, CFPB blog posts, OCC bulletins, internal regulatory affairs pages, or partner bank updates. Custom sources are checked on the same schedule as default sources and trigger the same alert chain when content changes.

6. Analytics, Audit Trail & Reporting

Every platform action is logged with user identity, timestamp, and detail. The analytics dashboard gives compliance leadership real-time visibility into risk posture, and the full audit log is examiner-ready from day one.

Analytics metrics

- **Submission volume** — by status, product, channel, date range
 - **Risk breakdown** — high / medium / low / pass across all submissions
 - **Top regulatory issues** — most-triggered frameworks and severity
 - **Conflict trends** — conflicts detected over time by product
- **False positive rate** — reviewer overrides as % of findings
 - **Review turnaround** — avg time submission → decision by reviewer
 - **Approval rate** — % approved vs. rejected vs. escalated
 - **Examiner readiness score** — composite risk posture indicator

Sample audit log entries

Action	User	Detail	Time (UTC)
Login	compliance@co.com	Session started	2025-03-15 09:14
Submit	marketing@co.com	Q2 Freedom Unlimited Email Campaign	2025-03-15 09:31
AI Analysis	system	9 regs · 3 findings · 1 conflict	2025-03-15 09:32
Review Decision	legal@co.com	Approved — minor edits required	2025-03-15 10:05
Reg Change	system	CFPB UDAAP Guidance content changed	2025-03-15 14:22
Memory Add	admin@co.com	Sapphire Preferred Rewards T&Cs; v3.1	2025-03-15 15:10
Mark False Positive	compliance@co.com	Finding ID a3f9 — correct by policy v2.4	2025-03-15 16:44

Export options

- **Submissions JSON** — All submissions — status, submitter, product, timestamps
- **Analytics JSON** — Risk breakdown, top issues, conflict counts
- **Audit Log JSON** — Full action history — users, timestamps, detail
- **Findings DOCX** — Color-coded: yellow=changed, red=deleted, green=added
- **Batch Summary** — Multi-file results with per-file risk ratings

7. Developer Platform & Tech Stack

Every feature accessible via REST API. Four implementation flavors. Open source on GitHub — MIT license.

AI Engine ● Anthropic Claude (claude-sonnet-4-20250514) ● RAG via ChromaDB persistent vector store ● Chunked regulation indexing with metadata ● Context-injected prompts with citation extraction ● Separate collections: regulations, policies, agreements	Interfaces ● Streamlit web platform — 9-page application ● FastAPI REST API with auto-generated /docs ● Node.js / Express equivalent ● CLI: single-file check and batch processor ● Color-coded DOCX export via python-docx
Integrations ● Slack incoming webhooks (formatted blocks) ● SMTP email with HTML templates ● SQLite audit database (zero-config) ● Regulatory monitor daemon (configurable) ● Webhook-ready event system	Deployment ● Streamlit Cloud — one-click ● Docker / docker-compose ready ● FastAPI on AWS / GCP / Azure ● GitHub Actions CI included ● Full .env variable configuration

Key REST API endpoints

Endpoint	Description
POST /check/text	Check plain text — regulations + company memory
POST /check/file	Upload and check file (PDF, DOCX, TXT, image)
POST /check/full	Full check with conflict detection, unified result
POST /check/export	Check file → return color-coded DOCX in one call
POST /kb/ingest/url	Scrape URL and add to regulatory knowledge base
POST /memory/add/file	Add document to company prior communications store
GET /memory/stats	Company memory statistics by document type
DELETE /memory/document/{n}	Remove a document from company memory
GET /kb/retrieve?q=	Test retrieval — top matching regulation chunks
GET /analytics	Platform analytics summary

8. Quick Start Guide

From zero to running compliance checks in under five minutes.

Step 1 — Install & Configure

```
pip install -r requirements.txt
cp .env.example .env
# Edit .env — add your ANTHROPIC_API_KEY
```

Requires Python 3.10+. All dependencies in one requirements.txt. Chromadb, reportlab, anthropic, streamlit, fastapi all included.

Step 2 — Launch the Platform

```
streamlit run ui/app.py
# Opens at http://localhost:8501
# Sign in: compliance@company.com / password123
```

Four demo users seeded automatically on first run (admin, compliance, legal, marketing). All password: password123.

Step 3 — Seed the Knowledge Base

```
# Train Regulations tab → click preset buttons
# Loads CFPB Reg Z, Reg B, Reg V, FDCPA, UDAP, SCRA, SR 11-7
# Company Memory tab → Bulk load test documents
```

Or via CLI: `python -m monitor.reg_monitor --seed 'My Company'`

Step 4 — Submit Your First Document

```
# Sign in as marketing@company.com
# Click 'Submit Document'
# Paste any marketing copy or upload a PDF
# Select regulations and click 'Run Compliance Check'
```

Results appear in under 60 seconds with regulatory findings, conflict analysis, and remediation recommendations.

Demo accounts

Email	Password	Role	Access level
admin@company.com	password123	Admin	Full access — all features and settings
compliance@company.com	password123	Compliance	Review queue, decisions, analytics, audit log
legal@company.com	password123	Legal	Review queue, escalation, analytics, audit log
marketing@company.com	password123	Submitter	Submit documents, view own submission history

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